Course announcement: 254B, expansion in groups of Lie type

Terence Tao · 25 November 2011

https://terrytao.wordpress.com/2011/11/25/course-announcement-254b-expansion-in-groups-of-lie-type/

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In the Winter quarter (starting on January 9), I will be teaching a graduate course on expansion in groups of Lie type. This course will focus on constructions of expanding Cayley graphs on finite groups of Lie type (such as the special linear groups $SL_d(\mathbf{F}_q)$, or their simple quotients $PSL_d(\mathbf{F}_q)$, but also including more exotic "twisted" groups of Lie type, such as the Steinberg or Suzuki-Ree groups), including the "classical" constructions of Margulis and of Selberg, but also the more recent constructions of Bourgain-Gamburd and later authors (including some very recent work of Ben Green, Emmanuel Breuillard, Rob Guralnick, and myself which is nearing completion and which I plan to post about shortly). As usual, I plan to start posting lecture notes on this blog before the course begins.

Comments (4)

AADIL AMAN · 25 November 2011

Great, Keep posting the updates of the course…. I love graph theory!!! and I want to explore about cayley graphs and its recent uses… Your blog is seriously awesome!!
thanks a lot
-Aadil

Bogdan · 28 November 2011

Dear Tao,
My question is completely out of topic, but I do not see where to ask "general" questions. I was recently asked the following question: What is the probability that a random n to n symmetric matrix (say, ones on diagonal, upper diagonals entries are i.i.d, and entries below the diagonal are given by symmerty) is positive definite? Clearly, this probability goes to 0 with n, but how fast? I could not answer this, but I have impression that the question is more than classical and should be known… Could you please comment?

Terence Tao · 28 November 2011

I would recommend http://mathoverflow.net/ for general research-level maths questions if they are off-topic for the posts on this blog.

Bogdan · 29 November 2011

Ok, I know this site, I just wanted to ask namely you… Thanks anyway.